

Deep Sequential Architectures for Financial Time Series Classification

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Abstract—This report investigates whether deep sequential architectures improve short-horizon Bitcoin price direction classification relative to classical machine learning baselines. Minute-level Bitcoin market data were transformed into supervised learning samples using chronological splits, engineered technical indicators, and fixed-length historical windows. Future directional labels were defined using future log returns over a fixed prediction horizon, with threshold filtering applied to remove near-zero movements and reduce label noise. The study evaluated shallow baselines and a broad family of deep models, including multilayer perceptrons, convolutional models, recurrent networks, hybrid architectures, transformer-based models, and multi-task extensions. Performance was assessed using classification metrics, calibration metrics, and backtesting analysis. The results showed that although deep sequential models captured temporal structure and often produced more balanced predictions, they did not consistently outperform simpler baselines on out-of-sample data. These findings highlight the difficulty of extracting reliable predictive signal from short-horizon financial time series and support the view that high-frequency Bitcoin direction prediction remains a low signal-to-noise problem.

I. INTRODUCTION

Financial time series prediction remains one of the most challenging problems in machine learning due to the complex statistical properties of market data. Financial signals are inherently noisy, nonlinear, and non-stationary, with distributions that evolve over time and are influenced by many interacting factors. In high-frequency settings, such as minute-level trading data, these challenges become even more severe because the signal-to-noise ratio is extremely low while microstructure noise is strong. As a result, even small improvements in predictive performance are difficult to obtain and even harder to translate into robust trading strategies. Empirical studies have shown that short-horizon returns often contain limited predictable structure and are frequently dominated by noise [1].

From a machine learning perspective, this problem reflects the fundamental challenge of generalization in noisy environments. A model must learn patterns that minimize training loss while maintaining strong performance on unseen data. However, in financial time series, the underlying signal is often weak relative to noise, making models prone to overfitting. Increasing model complexity can improve in-sample performance but may degrade out-of-sample generalization if the model begins to capture noise rather than meaningful structure. This bias-complexity trade-off is especially critical in high-

frequency financial prediction, where even highly expressive models may fail to deliver consistent improvements.

Recent advances in deep learning have introduced powerful tools for modeling sequential data. Deep neural networks are capable of learning complex nonlinear representations and extracting features directly from raw inputs. In particular, sequential architectures are designed to capture temporal dependencies and interactions across time, which are central to financial data. These capabilities suggest that deep sequential models may provide advantages over classical machine learning approaches that treat observations independently.

Despite these advances, it remains unclear whether increasing model sophistication leads to meaningful improvements in financial prediction tasks. Deep architectures introduce additional challenges, including optimization difficulty, sensitivity to hyperparameters, and increased risk of overfitting. Furthermore, the non-stationary nature of financial markets may limit the ability of learned representations to generalize across time. This raises an important question about the practical value of deep learning in high-frequency financial settings.

This project investigates this question in the context of Bitcoin minute-level market data. Cryptocurrency markets provide a suitable testbed due to their high volatility, continuous trading, and availability of large-scale high-frequency data. The study evaluates a wide range of models under a unified experimental framework, including classical machine learning methods, shallow neural networks, deep sequential architectures, hybrid models, and multi-task learning formulations.

The central research question is:

How do different sequential deep learning architectures compare in learning temporal representations for short-horizon Bitcoin direction prediction?

The original hypothesis was that sequential deep architectures would outperform classical baselines by explicitly modeling temporal dependencies and capturing multi-scale patterns in financial data. Hybrid architectures and multi-task formulations were further expected to improve robustness by combining complementary modeling strategies and incorporating auxiliary learning signals. At the same time, the study aims to critically examine whether increased model complexity leads to meaningful out-of-sample improvement in a domain characterized by high noise and weak predictability.

The main contributions of this work are as follows. First, we provide a systematic comparison of classical machine learning

models and a diverse set of deep sequential architectures under a consistent experimental setup. Second, we analyze the effect of architectural design choices, including recurrence, convolution, and attention mechanisms, on temporal representation learning. Third, we evaluate model performance using both classification metrics and trading-oriented backtesting measures, providing a more comprehensive assessment of practical utility. Finally, we present empirical insights into the limitations of deep learning for high-frequency financial prediction, highlighting the gap between model complexity and real-world performance.

The remainder of this report describes the dataset construction, feature engineering pipeline, labeling strategy, model architectures, training procedures, and evaluation results. The findings are used to draw conclusions about the effectiveness of deep sequential models for financial time series prediction and to identify directions for future research.

II. RELATED WORK

Financial time series prediction has been extensively studied across statistical, machine learning, and deep learning frameworks [1]. Traditional approaches relied on models such as autoregressive (AR) and autoregressive moving average (ARMA) models [2], as well as volatility-based models such as GARCH [3], [4], which capture linear dependencies and volatility clustering in financial time series. While these models provide useful insights into market behavior, they are limited in their ability to represent nonlinear relationships and complex interactions present in modern financial data [1].

The introduction of machine learning methods expanded the modeling capability by enabling nonlinear feature learning. Models such as logistic regression, random forests, and gradient boosting have been widely applied to financial prediction tasks. These approaches demonstrated improved performance over classical statistical models when combined with engineered features, but their results in high-frequency settings have generally remained modest. This is largely because such models treat observations independently and do not explicitly model temporal dependencies, which are essential in sequential financial data [1].

Deep learning methods introduced architectures specifically designed to capture temporal structure. Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, were proposed to address the vanishing gradient problem and enable the modeling of long-term dependencies in sequences [5]. Gated Recurrent Units (GRU) further simplified this design while maintaining comparable performance [6]. In financial applications, these models demonstrated the ability to capture temporal patterns more effectively than classical methods; however, empirical studies have shown that their predictive improvements are often limited, especially for short-horizon forecasting tasks where noise dominates the signal [7].

Convolutional neural networks (CNNs) were later adapted for time series analysis by applying filters across temporal dimensions to detect local patterns. Temporal Convolutional

Networks (TCNs) extend this approach using dilated convolutions, enabling the capture of long-range dependencies while maintaining computational efficiency. Bai et al. showed that TCNs can outperform recurrent models across a range of sequence modeling benchmarks [8]. In financial contexts, CNN-based models have been effective at extracting short-term temporal features, but similar to RNNs, their performance gains are often constrained by the low signal-to-noise ratio inherent in financial data.

Hybrid architectures combining convolutional and recurrent components have been proposed to leverage both local feature extraction and sequential modeling. A notable example is DeepLOB, where convolutional layers are used to extract hierarchical representations from limit order book data, followed by recurrent layers to model temporal dependencies [9]. This approach achieved strong classification performance in predicting short-term price movements, highlighting the importance of structured input data such as order book information. However, such results are closely tied to the richness of the input representation and may not generalize to price-only data.

More recently, transformer-based models have introduced attention mechanisms as an alternative to recurrence. Vaswani et al. demonstrated that self-attention can effectively model long-range dependencies without sequential processing, leading to state-of-the-art performance in several domains [10]. In time series forecasting, the Temporal Fusion Transformer (TFT) integrates attention with gating mechanisms and variable selection networks, achieving strong performance in multi-horizon forecasting tasks and improving interpretability [11]. Despite these advances, the application of transformer models to financial prediction has produced mixed results, particularly when applied to noisy, high-frequency price data without additional contextual features.

From a theoretical perspective, financial prediction is constrained by the Efficient Market Hypothesis, which suggests that publicly available information is rapidly incorporated into asset prices. Lopez de Prado emphasized that financial machine learning requires careful handling of non-stationarity, overlapping samples, and data leakage to produce reliable results [1]. Empirical findings consistently indicate that short-horizon returns are difficult to predict due to their low signal-to-noise ratio, implying that increasing model complexity alone does not guarantee improved performance.

Overall, prior work suggests that while deep learning architectures offer powerful tools for modeling temporal data, their effectiveness in financial prediction is fundamentally limited by the characteristics of the data. Improvements in architecture often yield only marginal gains, particularly in high-frequency settings. This study builds on these insights by systematically evaluating a wide range of classical, deep, hybrid, and multi-task models within a unified experimental framework to better understand the practical limits of predictive performance in short-horizon Bitcoin direction classification.

III. DATASET AND DATA PREPARATION

A. Data Source

This study utilized high-frequency Bitcoin market data sampled at one-minute intervals. Each observation consists of the standard OHLCV variables, namely open, high, low, close, and traded volume [12]. These variables provide a complete representation of price dynamics and market activity at each time step.

The dataset spans multiple years of continuous trading, resulting in a large number of observations suitable for both classical machine learning models and deep sequential architectures. All subsequent analysis began from these raw OHLCV inputs, and no external data sources or auxiliary signals were incorporated. This ensured that the predictive task remained based solely on historical market behavior.

B. Chronological Splitting

To ensure a realistic evaluation of model performance, the dataset was divided into training, validation, and test sets using strict chronological ordering. The earliest portion of the data was assigned to the training set, followed sequentially by the validation and test sets.

This approach preserves the temporal structure of financial data and prevents look-ahead bias. Since future information must not influence past predictions, chronological splitting ensures that models are evaluated on data that occur strictly after the training period.

In addition, a purge gap equal to the prediction horizon H was introduced between dataset partitions. Because the target variable depends on future prices over this horizon, adjacent samples near split boundaries may share overlapping information. Removing these boundary samples reduces leakage and helps ensure that training and evaluation sets remain statistically independent.

C. Target Construction

The prediction problem was formulated as a short-horizon binary direction classification task. The target variable was defined using the future log return over a fixed horizon H :

$$r_{t,H} = \log \left(\frac{C_{t+H}}{C_t} \right) \quad (1)$$

where C_t denotes the closing price at time t , and C_{t+H} represents the closing price after H future time steps. The use of log returns provides a symmetric and scale-stable representation of price movement.

To reduce the influence of noise associated with very small price fluctuations, a threshold filtering criterion was applied. Only samples satisfying

$$|r_{t,H}| > \tau \quad (2)$$

were retained, where τ is a predefined return threshold. This filtering removes near-zero movements that are difficult to predict and do not contribute meaningful learning signals.

The binary label was then defined as

$$y_t = \begin{cases} 1, & r_{t,H} > 0 \\ 0, & r_{t,H} < 0 \end{cases} \quad (3)$$

after threshold filtering. As a result, the dataset focuses on significant upward and downward movements, yielding a more balanced and informative classification problem.

D. Feature Engineering

In addition to the raw OHLCV variables, a set of engineered features was constructed to capture key characteristics of market behavior, including momentum, volatility, and trend dynamics.

Momentum features were derived from returns computed over multiple lookback horizons, using both simple and logarithmic formulations. Volatility was characterized using rolling statistics such as standard deviation and price range measures. Trend-related information was captured through moving averages and their relative differences, allowing the representation of directional tendencies in the data.

Furthermore, commonly used technical indicators were included, such as the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Bollinger Band features, including normalized price position and band width. These indicators provide compact summaries of market conditions and are widely used in quantitative analysis.

All engineered variables were combined into a unified feature vector at each time step. Prior to model training, feature values were normalized using statistics computed from the training set. This normalization ensures numerical stability and prevents information leakage from validation or test data. This engineered feature representation served as the common input space for both classical machine learning models and deep sequence architectures, enabling a consistent comparison across model families.

E. Input Construction

For sequential models, the data were transformed into fixed-length sequences of engineered feature vectors rather than raw price series. Each timestep is represented by a multidimensional feature vector containing OHLCV-derived indicators, momentum features, volatility measures, and trend-related signals. Each sample consists of a sliding window of length L :

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t] \quad (4)$$

where each $x_t \in \mathbb{R}^d$ denotes the engineered feature vector at time t . The corresponding label y_t is aligned with the final timestep of the input window.

For baseline models, each historical window was converted into a fixed-length tabular representation using statistical summaries of the same engineered features over the same lookback interval. These summaries included quantities such as the mean, standard deviation, minimum, maximum, and final observed value within each window. This produced a fixed-length representation suitable for classical machine learning

models while preserving comparability with the sequence-based inputs.

IV. METHODOLOGY

A. Sequence Representation

The predictive models in this study operated on fixed-length historical windows constructed from engineered feature vectors. For sequence-based models, each input sample was represented as

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t] \quad (5)$$

where each $x_t \in \mathbb{R}^d$ denotes the engineered feature vector at time t , L is the sequence length, and d is the feature dimension. The target label was aligned with the final timestep of the input window. This representation allows each model to learn temporal structure from a multivariate sequence rather than from a single observation.

Sequence windows were generated using stride-based sampling. This reduced redundancy between neighboring samples while preserving chronological order. The same sequence construction logic was applied to the training, validation, and test sets, with split-specific stride settings used where appropriate. In all cases, sequence generation respected dataset boundaries established during chronological splitting, so no sequence crossed a purge-gap boundary or leaked information across partitions.

B. Baseline Tabular Representation

For classical machine learning baselines, each historical window was converted into a fixed-length tabular representation using statistical summaries of the engineered features over the same lookback interval. These summaries included descriptive quantities such as the mean, standard deviation, minimum, maximum, and final observed value within each window. This design ensured that baseline models and deep sequential models were trained on representations derived from the same underlying information, enabling a controlled comparison across model families.

C. Single-Task Model Families

The first stage of the study focused on single-task binary direction classification. The evaluated models covered a broad range of complexity levels.

Classical baselines included Logistic Regression, Random Forest, and XGBoost. A multilayer perceptron operating on window-based summary features was also included as a neural baseline.

The deep sequential model zoo was designed to capture temporal structure in different ways. One-dimensional convolutional models were used to detect local temporal motifs. Recurrent architectures, including GRU and LSTM, were used to model sequential dependencies through hidden-state propagation. Hybrid CNN-GRU architectures combined convolutional feature extraction with recurrent temporal modeling. Transformer-based encoders and CNN-Transformer hybrid

models were further evaluated to determine whether attention-based sequence modeling could improve temporal representation learning in this setting.

D. Multi-Task Learning Extension

To investigate whether auxiliary supervision could improve the learned temporal representation, the study extended the single-task architecture into a multi-task framework. In addition to the primary binary direction classification objective, two auxiliary tasks were introduced: regime classification and future return regression.

Under this formulation, a shared sequence encoder produced a latent representation that was passed to separate task-specific heads. The direction head produced logits for the binary classification task, the regime head predicted the market regime, and the regression head estimated future return. The intuition behind this design is that jointly learning direction, regime, and return information may encourage the shared encoder to capture broader and more robust market structure.

The total multitask objective was defined as

$$\mathcal{L} = \lambda_{\text{dir}}\mathcal{L}_{\text{dir}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{ret}}\mathcal{L}_{\text{ret}} \quad (6)$$

where $\mathcal{L}_{\text{dir}}$ denotes the direction classification loss, $\mathcal{L}_{\text{reg}}$ denotes the regime classification loss, and $\mathcal{L}_{\text{ret}}$ denotes the regression loss for future return. The coefficients λ_{dir} , λ_{reg} , and λ_{ret} controlled the relative contribution of each task during training.

V. TRAINING PROCEDURE

All neural models were trained using gradient-based optimization with AdamW. For the primary direction classification task, the learning objective was based on cross-entropy loss. Class weighting was used where appropriate to reduce the effect of residual imbalance in the training data. In the multitask setting, the overall loss was computed as the weighted combination of direction classification loss, regime classification loss, and return regression loss.

To improve training stability and generalization, regularization strategies including dropout, weight decay, and early stopping were employed. Gradient clipping was applied to mitigate unstable updates in deeper sequence models. A OneCycle learning rate scheduler was used to improve convergence behavior across training epochs. Mixed precision training was also incorporated when available in order to improve computational efficiency.

A fixed classification threshold of 0.5 was used during final evaluation. This decision ensured consistency across models and avoided bias introduced by post hoc threshold tuning.

VI. EVALUATION METRICS

Model performance was evaluated using both discrimination and calibration metrics. Classification quality was measured using accuracy, balanced accuracy, macro F1 score, and weighted F1 score. Balanced accuracy was especially important because it provides a more reliable measure of

predictive quality when class frequencies are not perfectly symmetric.

In addition to classification performance, calibration quality was assessed using the Brier score and expected calibration error (ECE). These metrics were used to determine whether the predicted probabilities were well aligned with observed outcomes.

Finally, the study used a unified comparison framework in which classical baselines, deep single-task models, and multitask models were evaluated under the same data partitions, label construction, and metric definitions. This made it possible to compare predictive behavior across model families in a controlled setting. To assess practical usefulness, a backtesting procedure translated model predictions into trading signals under a simplified event-style trading framework, allowing the study to examine whether improvements in machine learning metrics translated into practical financial relevance.

VII. RESULTS

A. Overall Model Performance

The final experiments showed that short-horizon Bitcoin direction prediction remained difficult across all model families. Although a broad range of models was evaluated, including classical baselines, shallow neural networks, deep single-task models, and multi-task extensions, the observed performance differences were modest and remained close to chance level. This outcome is consistent with the weak predictive structure expected in high-frequency financial data.

Table I presents the combined single-task leaderboard across classical baselines and deep models. The strongest overall single-task result was obtained by the GRU model, which achieved an accuracy of 0.5157, balanced accuracy of 0.5158, macro F1 score of 0.5155, and weighted F1 score of 0.5155. The best baseline was Random Forest, which achieved an accuracy of 0.5150 and balanced accuracy of 0.5142. This is an important result because it shows that a strong tree-based baseline remained highly competitive with more complex deep sequential architectures.

Among the remaining single-task deep models, CNN-GRU ranked second overall, followed by TransformerEncoder, MLP(sequence), LSTM, CNN-Transformer-Attn, and CNN1D. At the baseline level, MLP(summary), XGBoost, and Logistic Regression all performed slightly below the strongest recurrent and tree-based models. However, the absolute differences between models remained small, reinforcing the conclusion that model choice alone did not overcome the weak-signal environment.

To assess whether auxiliary supervision improved the main direction prediction task, the study also compared single-task and multitask models under the same data partitions, feature space, and evaluation protocol. As shown in Table II, multitask learning did not outperform the strongest single-task model. The best overall result remained the single-task GRU, while the strongest multitask result was obtained by MultiTaskCNNTransformerAttention, which achieved an

TABLE I
COMBINED SINGLE-TASK LEADERBOARD ON THE TEST SET.

Model	Accuracy	Bal. Acc.	Macro F1	Weighted F1
GRU	0.5157	0.5158	0.5155	0.5155
RandomForest	0.5150	0.5142	0.5021	0.5025
CNN-GRU	0.5143	0.5139	0.5110	0.5112
TransformerEncoder	0.5127	0.5127	0.5126	0.5126
MLP(sequence)	0.5127	0.5122	0.5084	0.5087
LSTM	0.5130	0.5121	0.4982	0.4986
CNN-Transformer-Attn	0.5129	0.5120	0.4979	0.4983
CNN1D	0.5132	0.5120	0.4850	0.4856
MLP(summary)	0.5093	0.5102	0.4919	0.4914
XGBoost	0.5106	0.5098	0.4970	0.4974
LogisticRegression	0.5091	0.5088	0.5077	0.5078

TABLE II
COMPARISON BETWEEN SINGLE-TASK AND MULTITASK MODELS ON THE TEST SET.

Model	Accuracy	Bal. Acc.	Macro F1	Weighted F1	Type
GRU	0.5157	0.5158	0.5155	0.5155	Single
MultiTaskCNNTransformerAttention	0.5138	0.5142	0.5119	0.5118	Multi
CNN-GRU	0.5143	0.5139	0.5110	0.5112	Single
MultiTaskLSTM	0.5139	0.5138	0.5135	0.5136	Multi
MultiTaskCNNGRU	0.5140	0.5137	0.5127	0.5128	Multi
MultiTaskGRU	0.5141	0.5136	0.5090	0.5093	Multi
TransformerEncoder	0.5127	0.5127	0.5126	0.5126	Single
MultiTaskCNN1D	0.5130	0.5124	0.5063	0.5066	Multi
MultiTaskTransformerEncoder	0.5130	0.5123	0.5047	0.5050	Multi
MLP(sequence)	0.5127	0.5122	0.5084	0.5087	Single
LSTM	0.5130	0.5121	0.4982	0.4986	Single
CNN-Transformer-Attn	0.5129	0.5120	0.4979	0.4983	Single
CNN1D	0.5132	0.5120	0.4850	0.4856	Single

accuracy of 0.5138, balanced accuracy of 0.5142, macro F1 score of 0.5119, and weighted F1 score of 0.5118.

Several multitask models nevertheless remained competitive with the stronger single-task architectures. MultiTaskLSTM, MultiTaskCNNGRU, and MultiTaskGRU all achieved balanced accuracy values above 0.5136, placing them close to CNN-GRU and above several weaker single-task models such as LSTM, CNN-Transformer-Attn, and CNN1D. This suggests that multitask learning provided some benefit for selected architectures, but the improvement was not large enough to alter the overall ranking of the best models.

B. Calibration and Confidence Analysis

In addition to classification performance, the models were evaluated using calibration and confidence-based diagnostics. These analyses are important because small differences in accuracy or balanced accuracy do not fully describe how a model behaves probabilistically. In a weak-signal financial setting, the sharpness and reliability of predicted probabilities are often as informative as the final classification metric.

The strongest single-task deep model, GRU, produced an accuracy of 0.5157, balanced accuracy of 0.5158, macro F1 score of 0.5155, and weighted F1 score of 0.5155 on the test set. At a fixed classification threshold of 0.5, the model generated 36,779 downward predictions and 40,306 upward predictions. The resulting confusion matrix showed that both classes were predicted with only a slight edge above chance, confirming that the model had learned a weak but nontrivial discriminative pattern rather than a strong directional rule. Its

calibration quality was also reasonable, with a Brier score of 0.2497 and an expected calibration error of 0.0076.

The probability histogram for GRU, shown in Fig. 1, indicates that the predicted probabilities were concentrated around the decision boundary rather than strongly polarized toward 0 or 1. This pattern is consistent with a difficult near-balanced binary task in which the model rarely identifies extremely confident opportunities. The confidence curve for GRU in Fig. 2 shows that higher-confidence subsets achieved somewhat stronger accuracy and balanced accuracy than the full test set, but the gain was modest. Thus, the model did exhibit some ability to rank predictions by reliability, but not enough to reveal a strongly separable high-confidence subset.

The best baseline model, Random Forest, displayed a similar pattern. Its probability distribution, shown in Fig. 3, remained concentrated in a relatively narrow interval around 0.5, with limited mass in the extreme tails. The corresponding confidence curve in Fig. 4 shows that the most confident predictions were somewhat stronger than the full set, but the improvement again was limited. This supports the broader conclusion that even the strongest baseline model operated in a weak-signal regime. In other words, Random Forest was able to identify slightly more reliable predictions at the top of its confidence ranking, but not enough to imply a large hidden edge.

The best multitask classifier, MultiTaskCNNTransformerAttention, exhibited a closely related behavior. Its probability histogram in Fig. 5 remained centered near the decision threshold, indicating limited predictive separation. The confidence curve in Fig. 6 showed only a modest decline from the most confident subset to the full test set. This suggests that multitask learning produced competitive but still cautious probability estimates. The multitask formulation therefore improved robustness for some architectures without transforming the task into one with strong class separation.

Taken together, the calibration and confidence results show that the models were not severely miscalibrated, but their probability outputs still reflected substantial uncertainty. The curves and histograms consistently indicate that the fundamental challenge of the task was weak predictive signal rather than unstable or pathological probability estimates. Confidence filtering helped slightly, but not enough to materially change the overall interpretation of the results.

C. Backtesting Analysis

To evaluate practical usefulness, the predicted class outputs were translated into trading decisions under a simplified event-style backtesting framework. Table III summarizes the combined backtest outcomes across baseline, single-task deep, and multitask models.

The backtesting results reveal an important distinction between predictive performance under machine learning metrics and realized performance under a trading-oriented evaluation. Although the GRU model achieved the strongest overall classification performance, it did not produce the strongest backtest result. Instead, the best trading outcome was obtained by the

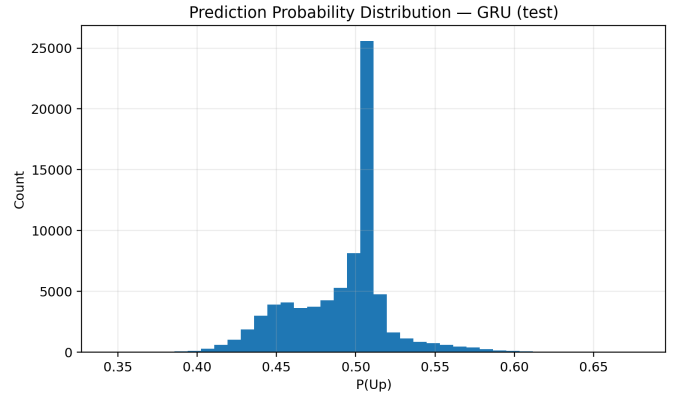


Fig. 1. Prediction probability distribution for the best single-task deep model, GRU, on the test set. The probabilities cluster near the decision threshold, reflecting limited class separability.

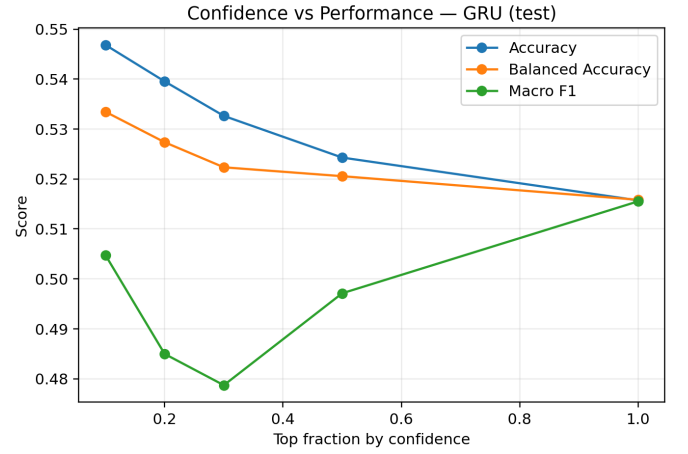


Fig. 2. Confidence-filtered performance for the best single-task deep model, GRU, on the test set. Higher-confidence subsets produce only modest improvement over the full test set.

LSTM model, which achieved a cumulative return of 4.2783, a Sharpe ratio of 2.3820, and a maximum drawdown of -0.2329. The GRU model ranked second in cumulative return at 2.8734, followed closely by CNN-GRU at 2.7617. Among the baseline models, Random Forest produced the strongest trading performance, with a cumulative return of 1.8561 and a Sharpe ratio of 1.5921.

These results show that the model with the best classification metrics does not necessarily produce the strongest trading performance. GRU achieved the highest balanced accuracy and macro F1 score among the single-task models, yet LSTM produced a substantially stronger cumulative return and Sharpe ratio under the backtesting framework. This discrepancy suggests that small differences in predicted class distribution, directional bias, and sequencing of correct versus incorrect trades can have a large effect on financial outcomes even when standard classification metrics are similar.

The baseline models further reinforce this point. Random Forest remained competitive not only in classification but also

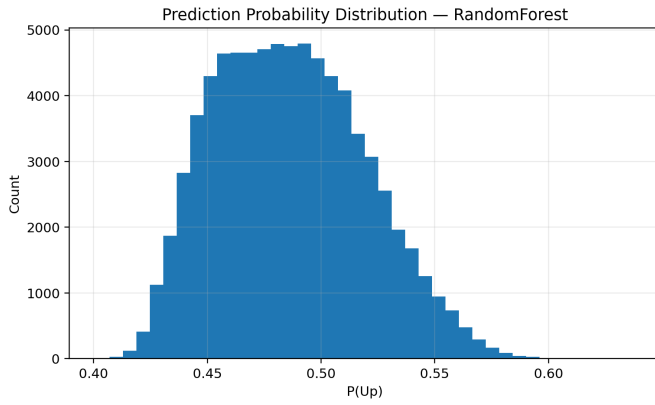


Fig. 3. Prediction probability distribution for the strongest baseline model, Random Forest. The narrow concentration around 0.5 indicates limited confidence in directional separation.

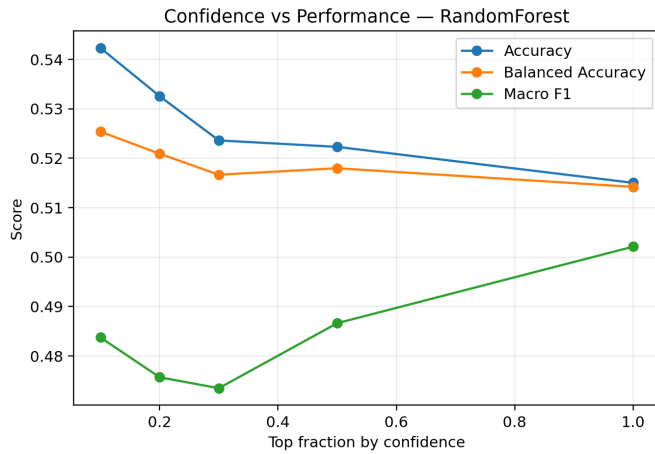


Fig. 4. Confidence-filtered performance for the strongest baseline model, Random Forest. Although the most confident predictions are somewhat stronger, the overall gain remains limited.

in backtesting, outperforming several deeper architectures in cumulative return. In contrast, XGBoost produced a negative cumulative return of -0.3023 and the weakest Sharpe ratio in the entire comparison. This indicates that modest classification performance is not sufficient by itself to guarantee trading usefulness, and that model behavior under sequential decision-making can differ substantially from its static classification ranking.

Among the multitask models, MultiTaskLSTM produced the strongest trading performance, with a cumulative return of 1.3859 and a Sharpe ratio of 1.3605. MultiTaskTransformerEncoder and MultiTaskCNNTransformerAttention followed, but none of the multitask models matched the strongest single-task recurrent models in financial performance. Although multitask learning produced competitive classification results, its trading outcomes remained modest relative to the best single-task sequence models.

Figure 7 shows the equity curve for the LSTM model, which produced the strongest cumulative return in the backtesting

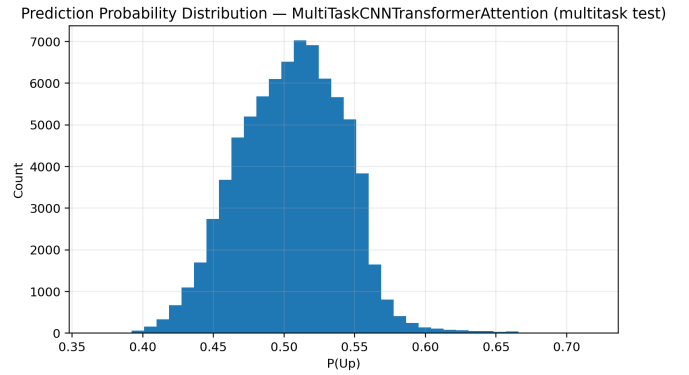


Fig. 5. Prediction probability distribution for the best multitask classifier, MultiTaskCNNTransformerAttention, on the test set. The distribution remains centered near the threshold, indicating weak predictive separation.

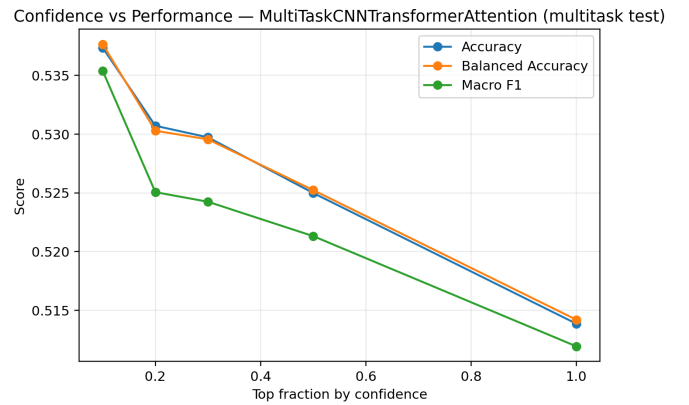


Fig. 6. Confidence-filtered performance for the best multitask classifier, MultiTaskCNNTransformerAttention, on the test set. Confidence filtering provides only limited improvement.

analysis. The curve demonstrates substantial growth over the trade sequence, but also includes noticeable intermediate drawdowns. This indicates that even the strongest backtesting model did not produce a perfectly smooth growth path. The equity axis is normalized, so an initial value of 1.0 corresponds to starting capital. Under this normalization, the final equity above 5 indicates more than a fivefold increase relative to the starting capital rather than a profit of only a few absolute dollars.

Overall, the backtesting results strengthen the central conclusion of the study. Weak but measurable classification edges were present, but these edges did not translate uniformly into financial performance. Some models converted small classification advantages into stronger cumulative returns, while others with similar predictive quality produced weaker or even negative trading outcomes. This highlights the importance of evaluating financial prediction models using both machine learning metrics and trading-oriented measures. In short-horizon Bitcoin prediction, modest improvements in balanced accuracy do not automatically imply practical financial advantage.

TABLE III
COMBINED BACKTESTING RESULTS ACROSS BASELINE, SINGLE-TASK,
AND MULTITASK MODELS.

Model	Cum. Return	Sharpe	Max Drawdown	Type
LSTM	4.2783	2.3820	-0.2329	Single-task
GRU	2.8734	1.9841	-0.2576	Single-task
CNN-GRU	2.7617	1.9461	-0.2792	Single-task
RandomForest	1.8561	1.5921	-0.2744	Baseline
MLP(sequence)	1.5559	1.4491	-0.2241	Single-task
MultiTaskLSTM	1.3859	1.3605	-0.3066	Multitask
MultiTaskTransformerEncoder	1.2925	1.3089	-0.2693	Multitask
MLP(summary)	1.2316	1.2750	-0.3937	Baseline
TransformerEncoder	1.1067	1.2005	-0.3037	Single-task
MultiTaskCNNTransformerAttention	0.9772	1.1190	-0.3100	Multitask
LogisticRegression	0.7336	0.9493	-0.3056	Baseline
CNN1D	0.7146	0.9357	-0.3585	Single-task
CNN-Transformer-Attn	0.6123	0.8567	-0.3176	Single-task
MultiTaskCNNGRU	0.4938	0.7584	-0.3052	Multitask
MultiTaskGRU	0.4724	0.7399	-0.2967	Multitask
MultiTaskCNN1D	0.2522	0.5315	-0.3113	Multitask
XGBoost	-0.3023	-0.2208	-0.5250	Baseline

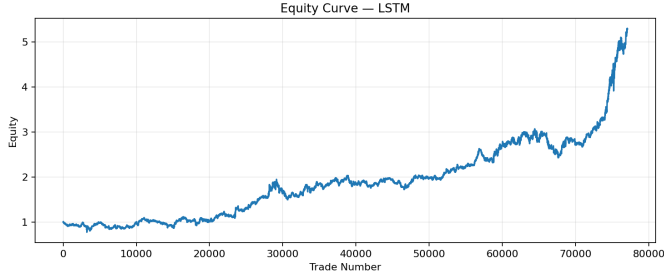


Fig. 7. Equity curve for the best backtesting model, LSTM, under the simplified event-style backtesting framework. The curve shows strong cumulative growth over the trade sequence, although several intermediate pullbacks remain visible.

VIII. DISCUSSION

The results of this study show that short-horizon Bitcoin direction prediction remains a difficult problem even when a wide range of modern deep learning architectures is considered. Although several models achieved performance slightly above chance, the overall gains were modest and highly compressed. This outcome suggests that the dominant limitation in this task is not the lack of model capacity, but the weak and unstable predictive signal available in minute-level price-based data.

One important finding is that recurrent architectures performed more consistently than the transformer-based and purely convolutional alternatives in the single-task setting. In particular, the GRU achieved the strongest overall classification performance, while CNN-GRU also ranked highly. This suggests that explicit temporal memory remained useful for modeling short-horizon price dynamics. At the same time, the absolute margin of improvement over the strongest baseline was small. Random Forest remained highly competitive and performed nearly as well as the best deep model. This is an important observation because it indicates that, in weak-signal settings, carefully constructed classical baselines can remain difficult to surpass even when more expressive sequential models are available.

The multi-task results provide a second important insight. The addition of auxiliary supervision through regime clas-

sification and future return regression did not outperform the strongest single-task model. Although the best multitask models remained competitive, the multitask formulation did not change the ranking of the top architectures. This suggests that auxiliary tasks may enrich shared internal representations without substantially improving the main binary direction objective. In other words, the shared encoder may learn broader market structure, but that information does not necessarily translate into stronger short-horizon directional predictability.

The calibration and confidence analyses reinforce this interpretation. The probability histograms for the strongest baseline, single-task, and multitask models were concentrated near the decision threshold, and the confidence curves showed only modest gains when evaluation was restricted to the most confident predictions. This indicates that the models were not severely miscalibrated, but they also were not discovering a sharply separable subset of highly reliable opportunities. Confidence filtering improved performance slightly, but not enough to materially change the conclusion of the study. The models therefore appeared to possess limited ability to rank their own reliability, which is consistent with a weak but nonzero signal environment.

A particularly important result of the study is the mismatch between machine learning metrics and backtesting performance. The GRU model achieved the best classification performance, yet the strongest backtesting result was obtained by the LSTM model. Similarly, Random Forest remained competitive in both classification and trading performance, while XGBoost performed poorly in backtesting despite non-catastrophic classification metrics. These results show that small differences in classification accuracy, balanced accuracy, or F1 score do not map directly to financial usefulness. A model may benefit from favorable trade sequencing, directional bias, or risk profile even if its aggregate classification metrics are not the strongest. This finding underscores the importance of evaluating financial prediction systems using both statistical and trading-oriented criteria.

The backtesting results should also be interpreted cautiously. Although several models produced positive cumulative returns under the simplified event-style framework, this does not imply that the same performance would necessarily hold in a live trading environment. Transaction costs, slippage, execution latency, market impact, and regime instability could materially reduce realized profitability. The backtesting framework was therefore useful for comparative analysis, but it should not be interpreted as a fully realistic trading simulation.

Several limitations of the study should be acknowledged. First, the predictive task was based entirely on historical OHLCV-derived information and engineered technical features. No order book data, trade flow data, or exogenous information sources were included. As a result, the input space may not have contained enough microstructure-level information to support stronger short-horizon predictability. Second, the target design focused on a binary direction label after threshold filtering. While this reduced label noise, it may also have simplified the market response into a form that does

not fully capture the underlying economic structure. Third, the experiments were evaluated on one specific market and one primary horizon, which limits the extent to which the findings can be generalized without further testing.

Despite these limitations, the study contributes an important practical insight. It demonstrates that increasing architectural sophistication alone is not sufficient to overcome the challenges of weak-signal financial prediction. In this setting, the differences between recurrent, convolutional, transformer-based, and multitask models were real but small. The broader conclusion is that model design matters, but data quality, target design, and evaluation framework matter at least as much. In high-frequency financial prediction, the main challenge may lie less in choosing ever more complex models and more in identifying more informative inputs and more meaningful prediction objectives.

IX. FUTURE WORK

Several directions remain open for improving the present study and extending its scope beyond the current experimental setting. The first and most immediate direction is additional training refinement. Although the current results already provide a clear comparative analysis, the models were trained under a practical but limited hyperparameter search process. Further work over the remaining project period can focus on more systematic optimization of learning rate schedules, dropout levels, weight decay, sequence length, stride selection, and class weighting. In particular, the recurrent models, which produced the strongest classification performance, may benefit from more targeted hyperparameter tuning and architecture-specific regularization.

A second direction is improved input representation. The current study used engineered features derived from OHLCV data, which capture momentum, trend, and volatility information but do not fully represent market microstructure. Future work should investigate richer inputs such as order book imbalance, bid-ask depth, trade flow, and other event-level variables. These signals are more closely aligned with short-horizon market behavior and may provide a stronger test of convolutional, recurrent, and hybrid architectures than OHLCV-derived features alone.

A third direction is architectural refinement guided by prior literature. The present study adopted a unified experimental framework in order to compare model families under consistent data preparation, training, and evaluation settings. Future work can extend this framework by incorporating additional architectural elements emphasized in prior studies, such as richer attention mechanisms, variable selection modules, gating structures, or input representations tailored to market microstructure data. These extensions would provide a more direct test of whether the current performance limits arise primarily from model design or from the information content of the input data.

A fourth direction is target reformulation. The current task used a threshold-filtered binary direction label in order to reduce noise from very small movements. While this was

appropriate for the present analysis, future work could explore alternative formulations such as multi-horizon prediction, return magnitude forecasting, volatility prediction, selective classification, or ranking-based objectives. These alternatives may better capture the economic structure of short-term market behavior and may be more suitable for trading decisions than unconditional binary classification alone.

A fifth direction is improved calibration-aware decision making. The confidence plots showed that the models possessed limited but non-negligible ability to rank their predictions by reliability. Future work should therefore examine selective trading rules in which only the highest-confidence predictions are acted upon, as well as calibration-aware thresholds that vary with market conditions. This may provide a more realistic path for converting weak classification edges into more robust trading outcomes.

A sixth direction is a more realistic trading evaluation framework. The current backtesting procedure was useful for model comparison, but it remained simplified relative to live market execution. Future work should incorporate explicit transaction costs, slippage, latency, execution constraints, and position sizing rules. It would also be valuable to evaluate the stability of trading performance across different market regimes rather than relying on a single aggregate result.

Finally, broader generalization should be tested. The present study focused on minute-level Bitcoin data, which represents a particularly noisy and difficult prediction setting. Future work could examine whether the observed conclusions hold across other assets, timeframes, and market structures. This includes comparing cryptocurrency markets across exchanges, as well as investigating whether lower-frequency datasets or richer market microstructure data produce a clearer advantage for deep sequential models.

Taken together, these directions suggest that the next phase of this project should move beyond simply adding model complexity. The strongest opportunities for improvement are likely to come from better input information, more carefully aligned prediction targets, stronger calibration-aware decision rules, and more realistic evaluation procedures.

X. CONCLUSION

This study examined whether deep sequential neural architectures improve short-horizon Bitcoin direction prediction relative to classical machine learning baselines. Using minute-level Bitcoin market data, engineered OHLCV-derived features, chronological data splits, and a threshold-filtered binary direction target, the work evaluated a broad set of models including classical baselines, single-task deep sequence models, hybrid architectures, and multitask extensions.

The results showed that the prediction task remained difficult across all model families. The strongest single-task classification result was obtained by the GRU model, while Random Forest remained the strongest classical baseline and highly competitive with the deep models. Multitask learning produced competitive performance for several architectures,

but it did not outperform the best single-task model. Calibration and confidence analyses further showed that most predicted probabilities remained concentrated near the decision threshold, indicating limited class separability despite reasonable calibration behavior.

Backtesting provided an additional practical perspective. The strongest classification model was not the strongest trading model, with LSTM producing the best backtesting outcome under the simplified evaluation framework. This result highlights the importance of evaluating financial prediction models not only with machine learning metrics, but also with trading-oriented measures.

Taken together, the findings indicate that increasing architectural complexity alone is not sufficient to overcome the weak predictive signal present in short-horizon Bitcoin price data. Future progress is more likely to depend on richer inputs, improved target design, and more realistic decision and evaluation frameworks than on model complexity alone.

A. Team Contributions

- **Gloria Mbaka:** Responsible for the full technical development of the project, including coding, model training, experimental design, analysis, and all written materials (proposal, report, and presentation).
- **Obumneme Umeonwuka:** Participated in the presentation.

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